

Kubernetes (K8s) Cheat Sheet

v1.30+ Ready

Quick Reference Guide: Architecture, kubectl, Workloads, Networking & Troubleshooting

1. CORE CONCEPTS & OBJECTS

Concept	Description
Pod	Smallest deployable unit (1+ containers sharing network/storage).
Deployment	Declarative updates for Pods & ReplicaSets; rolling updates.
Service	Stable network endpoint & load balancing for dynamic Pods.
Namespace	Virtual cluster partition for isolation and multi-tenancy.
ConfigMap	Stores plain non-sensitive key-value configuration data.
Secret	Stores sensitive data (passwords, tokens, keys) base64-encoded.
Ingress	Manages external HTTP/HTTPS routing to internal Services.
PV / PVC	PersistentVolume (storage) & PVC (workload storage claim).

2. ARCHITECTURE & COMPONENTS

Control Plane (Master):

- **kube-apiserver** : Cluster entry point
- **etcd** : Distributed key-value store
- **kube-scheduler** : Assigns Pods to Nodes
- **controller-manager** : State reconciler

Worker Node:

- **kubelet** : Node agent running Pods
- **kube-proxy** : Network packet filter/rules
- **container-runtime** : containerd / CRI-O

Flow: **kubectl** → **API Server** → **Scheduler/etcd** → **Kubelet** → **Container**

4. CORE YAML SPECS

Pod Manifest (pod.yaml)

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
  labels:
    app: web
spec:
  containers:
  - name: nginx
    image: nginx:alpine
    ports:
    - containerPort: 80
```

Deployment Manifest

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-dep
spec:
  replicas: 3
  selector:
    matchLabels:
      app: web
  template:
    metadata:
      labels:
        app: web
    spec:
      containers:
      - name: nginx
        image: nginx:latest
```

5. KUBERNETES SERVICES

Type	Use Case / Behavior
ClusterIP	Default. Internal cluster IP only (not reachable outside).
NodePort	Exposes service on static port (30000-32767) on each Node IP.
LoadBalancer	Provisions an external cloud Load Balancer (AWS ELB, GCP LB).
ExternalName	Maps service to an external CNAME record (no proxying).

Quick Expose: **kubectl expose deployment nginx-dep --port=80 --type=NodePort**

6. SCALING & ROLLING UPDATES

Action	Command
Scale replicas	kubectl scale deploy/nginx --replicas=5
Update Image	kubectl set image deploy/nginx nginx=nginx:1.25
Check Status	kubectl rollout status deploy/nginx
View History	kubectl rollout history deploy/nginx
Rollback	kubectl rollout undo deploy/nginx
Restart Pods	kubectl rollout restart deploy/nginx

7. CONFIGMAPS & SECRETS

Resource	Imperative Creation Command
ConfigMap	kubectl create cm app-cfg --from-literal=ENV=prod
Secret	kubectl create secret generic app-sec --from-literal=pass=123

View: **kubectl get cm** • **kubectl get secrets** • **kubectl get secret app-sec -o yaml**

3. ESSENTIAL KUBECTL COMMANDS

Command	Purpose
<code>kubectl cluster-info</code>	Display cluster details & health
<code>kubectl get nodes -o wide</code>	List nodes with IPs & OS details
<code>kubectl get pods -A</code>	List all pods across namespaces
<code>kubectl get all -n <ns></code>	List resources in specific namespace
<code>kubectl apply -f app.yaml</code>	Create/Update resources declaratively
<code>kubectl delete -f app.yaml</code>	Delete resources defined in file
<code>kubectl describe <type> <name></code>	Inspect detailed state & lifecycle events
<code>kubectl logs -f <pod> -c <container></code>	Stream container logs in real time
<code>kubectl exec -it <pod> -- /bin/sh</code>	Open interactive shell inside pod
<code>kubectl port-forward <pod> 8080:80</code>	Forward local port to Pod container

8. IMPORTANT POD STATES & HEALTH PROBES

State	Meaning & Cause
Pending	Waiting for scheduler or image pulling/volumes.
Running	Bound to a node, containers created & running.
CrashLoop	Container crashes repeatedly after startup.
ImagePull	Invalid image name, missing tag, or auth failure.

Health Probes:

livenessProbe	Checks if container is alive; restarts if failed.
readinessProbe	Checks if ready for traffic; removes from svc if failed.
startupProbe	Protects slow-starting containers before liveness starts.

9. STORAGE: VOLUMES, PV & PVC

- **Volume:** Tied to Pod lifecycle (e.g., `emptyDir`, `configMap`).
- **PV (PersistentVolume):** Provisioned storage in cluster.
- **PVC (Claim):** Workload request for specific storage size/access mode.
- **StorageClass:** Enables dynamic volume provisioning on demand.

`kubectl get pv,pvc` • `kubectl describe pvc <claim-name>`

10. TROUBLESHOOTING PLAYBOOK

Problem	First Diagnostic Commands
Pod Not Starting	<code>kubectl describe pod <pod></code> (check Events)
App Crashing	<code>kubectl logs <pod> --previous</code>
Service Unreachable	<code>kubectl get endpoints <svc></code> • verify labels
Node NotReady	<code>kubectl describe node <node></code> • check disk/mem
Recent Failures	<code>kubectl get events --sort-by=.lastTimestamp</code>

11. LABELS, SELECTORS & CONTEXTS

Action	Command
Show Pod Labels	<code>kubectl get pods --show-labels</code>
Filter by Label	<code>kubectl get pods -l app=nginx,env=prod</code>
Create Namespace	<code>kubectl create ns dev</code>
Set Default NS	<code>kubectl config set-context --current --namespace=dev</code>
Switch Context/Cluster	<code>kubectl config use-context <cluster-name></code>

12. CI/CD & MENTAL MODEL

Workflow: Code → Build Docker Image → Push to Registry → Write/Apply K8s YAML → Controller Schedules Pods → Service Routes Traffic → Monitor / Auto-scale.